GROUP – 16

PROJECT PROPOSAL

Universal Non Destructive Robotic Analog Automation System For Analog Audio Mixers

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INTRODUCTION

Audio mixing in live shows usually demands real time manual control adjustments by the sound technician to keep the sound levels in balance. When multiple channels are active, a technician must ride the faders continuously to prevent feedback and maintain a consistent mix volume which tends to make human errors.
While digital mixers with built in auto mixing features exist, their high cost makes them inaccessible for many users.
Our solution introduces a robotic automation system designed to make any analog audio mixer with auto-mixing capabilities. The system monitors input levels across channels and automatically adjusts faders and gain controls in real time, ensuring consistent sound balance while reducing the need for continuous manual intervention.

Background & Literature Review

Audio mixers play a crucial role in sound engineering, as they help in mixing and adjusting multiple sound signals and producing a sound output. Although analog mixers help in adjusting sound signals manually by adjusting volume and equalizer settings, it is a time consuming process and depends on the sound engineer's skills and expertise. Analog audio mixers require human intervention and precise adjustments, especially in dynamic environments such as live performances [2].

Recently, audio mixers have incorporated wireless technology and smart technologies such as mobile application control and speech recognition for easier control and operation. These systems still fundamentally require human intervention and manual input [2].

Parallel to these hardware improvements, the automation of music production tasks has emerged as a growing field in software engineering, with researchers exploring machine learning and deep learning approaches to perform automatic music mixing [1].However, the performance of these supervised deep learning models is heavily limited by a lack of "dry" instrument recordings available for training. As a result, purely software based automatic mixing is often still far from achieving the quality of professional human made mixes.[1]

To overcome this limitation, it is important to develop an automated audio mixer system that is able to adjust its parameters in real time and minimize human intervention. It is possible to achieve this by using sensors and actuators that can help in monitoring various environmental and audio conditions, such as sound levels, noise and adjust the mixer accordingly. It is a highly suitable application for smart audio systems since it is able to minimize human intervention and improve sound quality.

Objectives

The primary objective of this project is to design and implement a robotic, automated fader and gain control attachment for standard analog mixing consoles. To achieve this, the project encompasses the following specific goals

- Develop and calibrate an analog peak detector circuit to condition raw, fluctuating AC microphone signals into smooth, measurable DC voltages for the LabVIEW DAQ.
- Utilize a closed loop control architecture to calculate required acoustic corrections based on real time microphone/audio inputs and physically actuate faders to maintain a steady output target

- Implement a software based Noise Gate within LabVIEW to automatically mute inactive channels and prevent the amplification of ambient room noise.
- Utilize the real time envelope tracking to instantly detect sudden, dangerous spikes in audio amplitude characteristic of acoustic feedback between microphones and speakers and rapidly attenuate the gain and fader to break the feedback loop before it causes hardware damage or audience discomfort.

Scope

The scope of this project encompasses the design of custom signal conditioning circuitry (envelope detectors) to interface with microphone and other inputs, the mechanical fabrication of a motorized gantry system, and the development of a LabVIEW software architecture to process sensor data and execute closed loop control logic. Fader control will done by a motorized gantry system and gain knobs will control by servo motors. The system will not actuate Equalization (EQ) or Panoramic (Pan) controls, as these are treated as static artistic choices. Additionally a web base application will be developed to control the parameters.

Methodology

Hardware components, including sensors (TOF, SOI, Audio Envelop Sensor) and actuators (servo motors, stepper Motors for motorized gantry system) will be integrated DAQ card. Software development will be involved to edit parameters remotely. Observe the parameters that are marked on the fader of figure 01 .Each channels of the mixer will be included a TOF sensor to measure the X distance, SOI sensor to detect the fader when it is at B position and a audio envelop sensor to detect the peak voltage of the input audio signal which get directly from the audio input parallelly. The audio envelop sensor will operate as Figure 02.

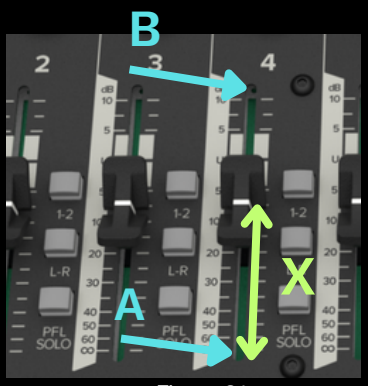


Figure 01

First a sound check will be done and the faders and gain levels will adjust to the suitable level manually. During this sound check the average peak voltage will be detected by the sensors for each channel (Set Voltage For each channel).Then the procedure follows as below for a one channel.

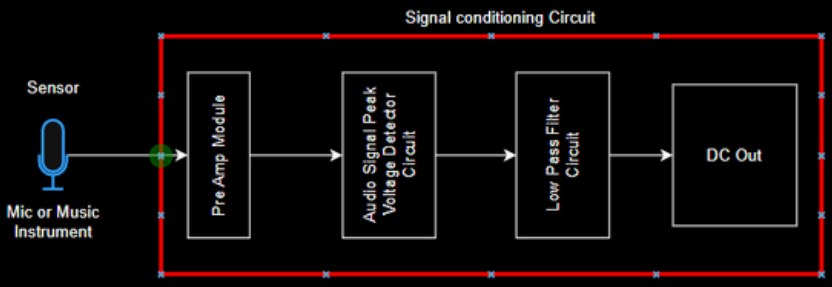
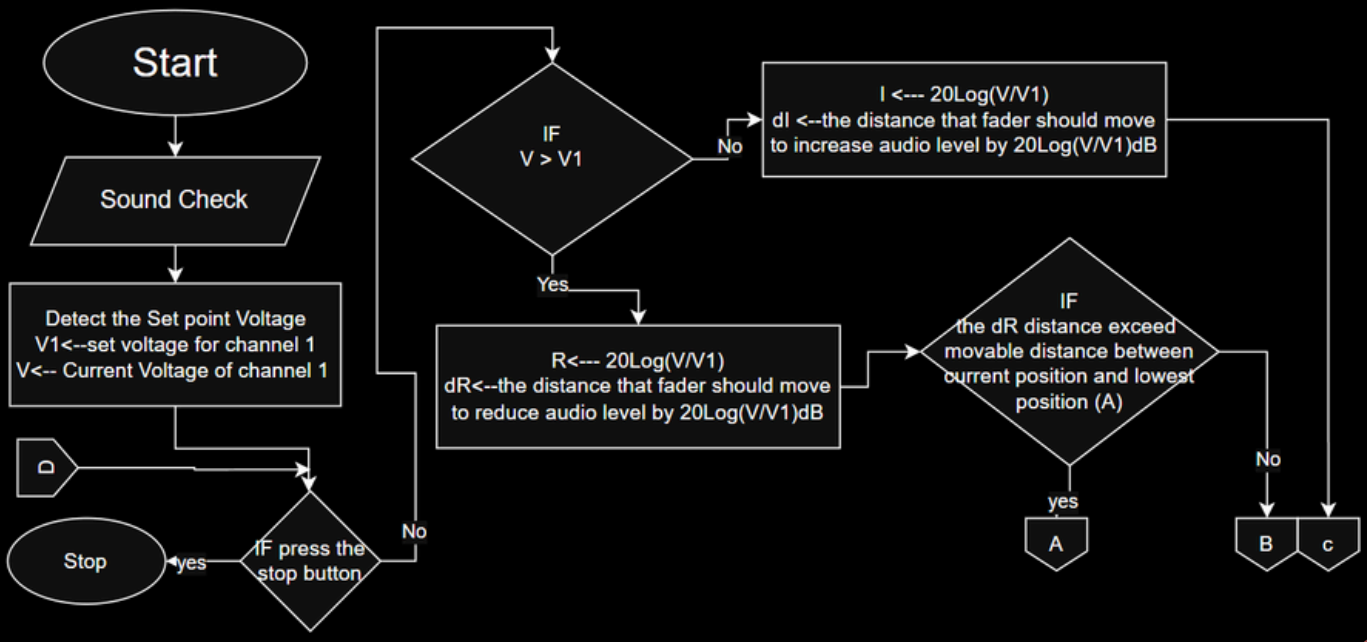
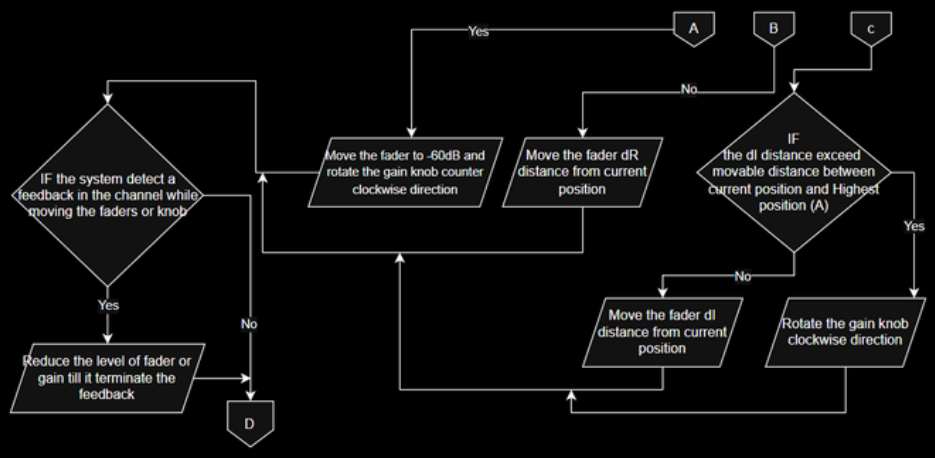


Figure 02





Logarithmic potentiometers are the industry standard for audio attenuation due to their exponential resistance characteristics [3]. However, for digital tracking and motor control, a linear reference is required. Our solution involves discretizing the exponential curve into linear divisions [4]. This mapping facilitates the precise detection of fader displacement and gain rotation, allowing the control algorithm to translate desired decibel changes into accurate mechanical movement

Budget

Made for 3 channels

Component	Price(Rs.)
TOF(VL53LOX) x 3	2670
SLOT(ITR9608) x 3	240
AD828 x 3	1200
Audio Peak Detector Circuit x 3	600

Low Pass Filters x 3	450
Stepper Motors x 6	3600
Belts and Pulleys x 4	3000
3D Printing	3000
	14760/=

Timeline

Conclusion

This project proposes a non-destructive robotic automation system for analog audio mixers using sensors, actuators, DAQ, and LabVIEW. The system enables real-time monitoring and automatic adjustment of audio parameters, reducing manual effort and improving sound consistency. It provides a cost-effective solution for enhancing analog mixer functionality and supports future advancements in smart audio automation.

Refferences

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